INCH-POUND
MIL-DTL-32688
1 October 2021

DETAIL SPECIFICATION

TROUSERS, TROPICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the United States Marine Corps Tropical Combat Uniform (MCTCU) trousers.

1.2 Classification. The MCTCU trousers are of one type in the following classes, sizes, and lengths as specified (see 6.2).

1.2.1 Classes.

Class 1 – Woodland MCTCU

Class 2 – Desert MCTCU

1.2.2 Schedule of sizes. The trousers are to be constructed in the sizes and lengths indicated in table I and table II.

TABLE I. Schedule of alpha sizes.

Sizes	X-Small	Small	Medium	Large	X-Large	XX-Large
Lengths	X-Short	X-Short	X-Short	X-Short	-	-
	Short	Short	Short	Short	Short	-
	Regular	Regular	Regular	Regular	Regular	Regular
	Long	Long	Long	Long	Long	Long
	-	X-Long	X-Long	X-Long	X-Long	X-Long
	-	-	XX-Long	-	-	XX-Long

TABLE II. Schedule of numeric sizes.

Sizes	24	28	32
Lengths	X-Short	X-Short	-
	Short	Short	Short
	-	Regular	-

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50198	-	Thread, Gimp, Cotton, Buttonhole
A-A-50199	-	Thread, Polyester Core, Cotton or Polyester-Covered
A-A-59826	-	Thread, Nylon
A-A-59994	-	Button, Sewing Hole, and Button, Staple, (Plastic)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-5664	-	Webbing, Textile, Elastic
MIL-DTL-32075	-	Label: For Clothing, Equipage, and Tentage, (General Use)
MIL-DTL-32439	-	Cloth, Duck, Textured Nylon
MIL-PRF-32659	-	Textiles: Vector Protection Treatment of
MIL-DTL-32679	-	Cloth, Tropical, Nylon and Cotton
MIL-C-44296	-	Cloth, Fusibles

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE LOGISTICS AGENCY (DLA) TROOP SUPPORT

DLA Troop Support Clothing & Textiles Additional Quality Assurance Requirements

(Copies of this document are available online at https://www.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9	-	Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM135	-	Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of this document are available online at www.aatcc.org.)

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AMERICAN SOCIETY FOR QUALITY

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D76/D76M - Standard Specification for Tensile Testing Machines for Textiles

ASTM D3990 - Standard Terminology Relating to Fabric Defects

ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at www.astm.org.)

CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 2014.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, public library, to name a few.)

SAE INTERNATIONAL

AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design. The MCTCU trousers shall be the United States Marine Corps design described in table III and as shown in figure 1 through figure 12. The trousers shall be constructed from basic material that has been treated with insect repellent in yardage form, or, as an alternate, the trousers may be constructed from untreated basic material and then treated with insect repellent in the garment form.

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TABLE III. Design.

Design Feature	Detail
Waist	Grown-on (standard) or sewn-on (alternate) waistband
	Partially elasticized waistband providing 4 inches of waist expansion (elasticized from front pocket bearer, across outseam, and onto back of trousers)
	Belt loops (7)
Fly	Concealed button and buttonhole (4) fly closure
Front pleats (2)	Single stitched, centered on each front panel
Side pockets (2)	Hanging
Hip pockets (2)	Hanging with welt opening
	Fused flap with concealed button closure
Cargo pockets (2)	Pleated (3) with a bottom bellow with a drainage eyelet at rear of pocket
	Elasticized upper pocket edge
	Fused flap closed with a bartack at forward corner and concealed button closure at rear corner
Crotch gusset	Stitched on
Reinforcement patches (3)	Single layer external patch at the seat and knees
Leg bottom	Hemmed
Fusible interfacing	Hip and cargo pocket flaps

3.4 Materials and components.

3.4.1 Treated basic material. The trousers shall be made of a lightweight nylon/cotton cloth that has been treated with an insect repellent. The basic material for the class 1 trousers shall conform to MIL-DTL-32679, type II, class 1 and the basic material for class 2 trousers shall conform to MIL-DTL-32679, type II, class 2 when tested as specified in 4.3.1.2. The camouflage pattern execution shall conform to the drawings referenced in MIL-DTL-32679 and as displayed by the standard sample.

3.4.1.1 Alternate untreated basic material. As an alternate, an untreated basic material conforming to MIL-DTL-32679, type I may be used to construct the trousers. After construction, the trousers shall be treated with an insect repellent in garment form in accordance with MIL-PRF-32659. After treatment, the alternate basic material for the class 1 trousers shall conform to MIL-DTL-32679, type II, class 1 and the alternate basic material for the class 2 trousers shall conform to MIL-DTL-32679, type II, class 2 when tested as specified in 4.3.2.2. The camouflage pattern execution shall conform to the drawings referenced in MIL-DTL-32679 and as displayed by the standard sample.

3.4.2 Reinforcement material. The reinforcement material for the knees and seat shall be a 330 denier, 4.0 ounce (untreated) textured nylon duck material conforming to the requirements in table IV when tested as specified in 4.3.1.2.

TABLE IV. Reinforcement material requirements.

Blouse	Requirement
Class 1	MIL-DTL-32439, type IV, class 1, style E (Woodland MARPAT)
Class 2	MIL-DTL-32439, type IV, class 1, style F (Desert MARPAT)

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3.4.3 Interfacing. The fusible interfacing for the hip and cargo pocket flaps shall conform to MIL-C-44296, type V, class 3, style B when tested as specified in 4.3.1.2. The fusible interfacing shall have the same life expectancy as the garment.

3.4.4 Thread. The threads shall conform to the requirements of table V when tested as specified in 4.3.1.2. The thread color for the type I trousers shall be Camouflage Green 483 or color chip 34094 of AMS-STD-595, and the thread color for the type II trousers shall be Khaki P-1or color chip 30277 of AMS-STD-595.

TABLE V. Thread requirements.

Thread	Location	Needle/Bobbin	Requirement	Size (Ticket/Tex)
A	High stress seams (see 3.8)	Needle	A-A-59826, type II, class B	B/50
		Bobbin (looper)	A-A-50199 1/	50/36 to 50
B	All bartacks and all other seams except high stress (see 3.8)	Needle and bobbin (looper)	A-A-50199, type I 1/	50/36 to 50
		Optional bobbin (looper)	A-A-50199, type I 2/	70/31 to 35
C	Button attachment	-	A-A-59826 type II, class A, size B	B/45 to 51
D	Buttonhole reinforcement	-	A-A-50198, type I	-/210

FOOTNOTES:

1/ Minimum breaking strength of 3.2 pounds.

2/ Minimum breaking strength of 2.6 pounds.

3.4.5 Elastic webbing. The elastic webbing shall conform to the requirements of table VI when tested as specified in 4.3.1.2.

TABLE VI. Elastic webbing.

Characteristic	Requirement
Physical requirements	MIL-W-5664, type II, class 1 (exception: force vs. elongation requirements as specified in 3.5)
Color	Black, white, or natural
Width	1½ inches
Length	Cut to a sufficient length to meet relaxed and stretched measurements of waistband and cargo pocket (see 3.9 and 3.9.1)
Ends	Heat cut to prevent raveling

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3.4.6 Buttons. The buttons shall conform to the requirements of table VII when tested as specified in 4.3.1.2.

TABLE VII. Button requirements.

Characteristic	Requirement
Physical requirements	A-A-59994, type I, class 1, style 26
Color for type I trousers	Color chip 24119 of AMS-STD-595 (Marine Olive Drab Green)
Color for type II trousers	Color chip 30340 of AMS-STD-595 (Tan 380)
Finish	Matte finish #21
Size	30 ligne

3.4.7 Labels. The trouser labels shall conform to the requirements of table VIII when tested as specified in 4.3.1.2. The permanent label inscription, legibility, label material, and label attachment method shall last the expected life of the uniform. Dimensional tolerance is ±⅛ inch. A sample of each label shall be submitted to the Government prior to use (see 6.2).

TABLE VIII. Labels.

Label Features	Requirements
1. Size label	
a. Physical requirements	(1) MIL-DTL-32075, type VI, class 2 and the following exceptions:
	(2) Dimensions: 1¾ inches wide by 1¼ inches long (excluding seam allowance).
	(3) Size label background color:
	(a) Class 1 trousers: Medium Green, Cable No. 70034, 70130, or 70131.
	(b) Class 2 trousers: Khaki, Cable No. 70188.
b. Inscription	(1) In accordance with figure 11.
	(2) 10 point Arial font, black.
	(3) Size and body measurements (waist and inseam) in accordance with table IX.
c. Location	(1) Caught in the waistband stitching with a ¼-inch seam allowance on the inside center back, left of the seat seam.
	(2) The stitching is not through the text.

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TABLE VIII. Labels – Continued.

Label Features	Requirements
2. Combination label (identification and instruction)	
a. Physical requirements	(1) MIL-DTL-32075, type VI, class 15 and the following exceptions:
	(2) Background color:
	(a) Class 1 trousers: Medium Green, Cable No. 70034, 70130, or 70131.
	(b) Class 2 trousers: Khaki, Cable No. 70188.
	(3) Dimensions: No larger than the pocket to which it is attached.
b. Inscription	(1) In accordance with figure 12.
	(2) Identification inscription:
	(a) For the class 1 trousers: TROUSERS, WOODLAND MARPAT™ CAMOUFLAGE MCTCU 50% Cotton/50% Nylon
	(b) For the class 2 trousers: TROUSERS, DESERT MARPAT™ CAMOUFLAGE MCTCU 50% Cotton/50% Nylon
c. Location	(1) Backside of the right hip pocket (as worn), with the inscription facing the wearer.
	(2) Sewn on all four sides.
	(3) The stitching is not through the text.
3-A. Insect protection label (including brand identification)	
a. Physical requirements	(1) MIL-PRF-32659.
	(2) No larger than the pocket to which it is attached.
b. Inscription	(1) MIL-PRF-32659.
c. Location	(1) Backside of the left hip pocket (as worn), with the inscription facing the wearer, in alignment with the combination label on the right hip pocket.
	(2) Sewn on all four sides.
	(3) The stitching is not through the text.
3-B. Alternate insect protection label with separate brand identification label	
a. Physical requirements	(1) Same as 3-A.a except the insect protection and the brand identification labels may be two separate labels.
b. Inscription	(1) MIL-PRF-32659.
c. Location	(1) Same requirements as label 3-A.c with the two labels in vertical alignment.
4. Insect protection hang tag	
a. Physical requirements	(1) MIL-PRF-32659.
b. Inscription	(1) MIL-PRF-32659.
c. Location	(1) Completely visible on the item when the item is folded or packaged as specified in 5.1.
	(2) Location and attachment method cause no damage to the item.

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TABLE VIII. Labels – Continued.

Label Features	Requirements
5. Lot designation	
a. Physical requirements	(1) DLA Troop Support Clothing & Textiles Additional Quality Assurance Requirements – Lot numbering procedure.
b. Inscription	(1) DLA Troop Support Clothing & Textiles Additional Quality Assurance Requirements – Lot numbering procedure.
c. Location	(1) Next to or above the combination label on the right hip pocket.
6. Barcode label	
a. Physical requirements	(1) MIL-DTL-32075, type VIII, class 17.
b. Inscription	(1) MIL-DTL-32075, type VIII, class 17.
c. Location	(1) Attached to the garment by a fastener.
	(2) Completely visible on the item when the item is folded or packaged as specified in 5.1.
	(3) Location and attachment method cause no damage to the item.

TABLE IX. Size and body measurements.

Characteristic	Size	Body Measurement (inches)
Alpha		
Waist (Size of trousers)	X-Small	Up to 27
	Small	27 to 31
	Medium	31 to 35
	Large	35 to 39
	X-Large	39 to 43
	XX-Large	Over 43
Inseam (Length of trousers)	X-Short	Up to 26½
	Short	26½ to 29½
	Regular	29½ to 32½
	Long	32½ to 35½
	X-Large	35½ to 38½
	XX-Large	Over 38½
Numeric		
Waist (Size of trousers)	24	22 to 26
	28	26 to 30
	32	30 to 34
Inseam (Lengths of trousers)	X-Short	Up to 26½
	Short	26½ to 29½
	Regular	29½ to 32½

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3.5 End item physical requirements. The trousers shall conform to the physical requirements specified in table X when tested as specified in 4.3.2.2.

TABLE X. End item physical requirements.

Characteristic	Requirement
Alternate basic material (after insect repellent treatment has been applied to garment)	
Class 1 trousers	MIL-DTL-32679, type II, class 1
Class 2 trousers	MIL-DTL-32679, type II, class 2
Garment (made from either the treated basic material or the alternate basic material that has been treated with the insect repellent in garment form)	
Dry out time (maximum): Medium-Regular size	5 hours
Weight of the trousers (maximum): Medium-Regular size	1 pound, 8 ounce
Button pull test (minimum)	40 pounds
Elastic force vs. elongation	
Cargo pocket (100 cycles)	2.5 to 3.5 pounds
Waistband at ¾-inch extension	0.5 to 1.5 pounds
Waistband at 1½-inch extension	1.5 to 4.0 pounds
Toxicity	See 3.5.1

3.5.1 Toxicity. The trousers shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.3.2.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.6 Figures. When inconsistencies exist between the written specification and figures 1 through 12, the written specification shall govern.

3.7 Patterns. Standard patterns will be furnished by the Government (see 6.2). Government patterns show size and directional lines for cutting and are marked or notched for the proper placement and assembly of component parts. The directional lines indicate the warp direction of the cloth unless otherwise indicated on the patterns. Patterns provide a seam allowance of ⅜ inch for single needle seams and ½ inch for double needle seams. The list of pattern parts and computer nomenclature is provided to ensure that the pattern set provided is complete. Additional notching is permitted if needed to facilitate manufacture. Minor modifications are permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment provided the modifications do not alter the dimensional, serviceability, or appearance requirements cited in this document or governed by the Government pattern.

3.7.1 Patterns when using treated basic material. When using the treated basic material (see 3.4.1), the Government patterns shall be utilized to manufacture the trousers, and the manufactured trousers shall meet the dimensional requirements of 3.9 and 3.9.1.

3.7.2 Patterns when using alternate untreated basic material. When using the alternate untreated basic material (see 3.4.1.1), the contractor shall prepare a set of working patterns by modifying the Government patterns to accommodate for the shrinkage of the trousers during the insect repellent application process. When specified (see 6.2), the contractor shall submit that working pattern file to the Government for review prior to cutting. Once approved, those working patterns shall be used to cut and manufacture trousers to meet the dimensional requirements of 3.9 and 3.9.1.

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3.7.3 List of trousers pattern parts. Component parts shall be cut from the materials and the number of parts specified in table XI and table XII. The pattern part lists in table XI and table XII are provided for first quality fabrics. When specified (see 6.2), off-shade basic material of the same basic material type may be used for bagged pockets only (i.e., non-visible areas). Two different sets of patterns are provided – one for alpha sizing (32688A) and one for numeric sizing (32688N).

TABLE XI. List of alpha sizing trouser pattern parts.

Material	Piece Name	Cut Parts
Basic material (see 3.4.1 and 3.7.1) or alternate basic material (see 3.4.1.1 and 3.7.2)	BACK	2
	FRONT	2
	LEFT FLY	3
	RIGHT FLY LINING	1
	HIP POCKET	2
	HIP POCKET FACING	2
	HIP POCKET BEARER	2
	HIP POCKET FLAP TAB	2
	SIDE HANGING POCKET	2
	SIDE HANGING POCKET BEARER	2
	SIDE HANGING POCKET FACING	2
	CARGO POCKET	2
	CARGO POCKET FACING	2
	CARGO POCKET FLAP TAB	2
	GUSSET	1
	WAISTBAND	1/
	BACK ALTERNATE	2
	FRONT ALTERNATE	2
	BUTTONHOLE FLY ALTERNATE	1
	HIP POCKET ALTERNATE	2
	LEFT FLY LINING ALTERNATE	1
	RIGHT FLY LINING ALTERNATE	1
	SIDE HANGING POCKET BEARER ALTERNATE	2
	SIDE HANGING POCKET FACING ALTERNATE	2
	SIDE HANGING POCKET ALTERNATE	2
Reinforcement material (see 3.4.2)	KNEE PATCH	2
	SEAT PATCH	2
Interfacing (fusible) (see 3.4.3)	CARGO POCKET FLAP TAB FUSIBLE	2
	HIP POCKET FLAP FUSIBLE	2

FOOTNOTE:

1/ Cut 1 for the grown-on waistband construction method or may be applied in roll form. Cut 2 for the sewn-on waistband construction method or may be applied in roll form.

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TABLE XII. List of numeric sizing trouser pattern parts.

Material	Piece Name	Cut Parts
Basic material (see 3.4.1 and 3.7.1) or alternate basic material (see 3.4.1.1 and 3.7.2)	BACK	2
	FRONT	2
	LEFT FLY	3
	RIGHT FLY LINING	1
	HIP POCKET	2
	HIP POCKET FACING	2
	HIP POCKET BEARER	2
	HIP POCKET FLAP TAB	2
	SIDE HANGING POCKET	2
	SIDE HANGING POCKET BEARER	2
	SIDE HANGING POCKET FACING	2
	CARGO POCKET	2
	CARGO POCKET FACING	2
	CARGO POCKET FLAP TAB	2
	GUSSET	1
	WAISTBAND	1/
	BACK ALTERNATE	2
	FRONT ALTERNATE	2
	BUTTONHOLE FLY ALTERNATE	1
	HIP POCKET ALTERNATE	2
	LEFT FLY LINING ALTERNATE	1
	RIGHT FLY LINING ALTERNATE	1
	SIDE HANGING POCKET BEARER ALTERNATE	2
	SIDE HANGING POCKET FACING ALTERNATE	2
	SIDE HANGING POCKET ALTERNATE	2
Reinforcement material (see 3.4.2)	KNEE PATCH	2
	SEAT PATCH	2
Interfacing (fusible) (see 3.4.3)	CARGO POCKET FLAP TAB FUSIBLE	2
	HIP POCKET FLAP FUSIBLE	2

FOOTNOTE:

1/ Cut 1 for the grown-on waistband construction method or may be applied in roll form. Cut 2 for the sewn-on waistband construction method or may be applied in roll form.

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3.8 Construction requirements. End item construction and appearance shall conform to the requirements in table XIII and table XIV, figure 1 through figure 12, and the finished dimensions cited in 3.9 and 3.9.1 to maintain end item configuration as verified in 4.3.4 and 4.3.5.

TABLE XIII. Construction details.

Construction Feature	Requirement
1. Thread	
a. Location	(1) Thread A (see table V) – high stress seams: trouser inseams, seat seams, seat patch, knee patch, and leg hem.
	(2) Thread B (see table V) – all other seams except high stress seams.
b. Tension	(1) Maintained so that there is no tight, loose, or unbalanced stitching.
c. Shade	(1) Consistent thread shade on the outside of the garment. Multiple shades of thread on the outside of the garment are not allowed.
d. Breaks (all stitch types)	(1) Secured by stitching back of the break not less than ¼-inch or ½-inch overlap.
	(2) If 401 stitch type, repaired using 301 stitch type.
e. Ends	(1) Trimmed to ¼ inch or less.
f. Loose	(1) All removed.
2. Stitches and seams	
a. Stitch and seam types	(1) In accordance with ASTM D6193.
	(2) In accordance with table XIV.
b. Appearance	(1) Consistent, exhibiting a uniform appearance.
c. Underside of seams (inside the garment)	(1) Flat, with no protruding seam allowances or raw edges that could cause irritation, discomfort, or poor appearance.
d. Stitches per inch	(1) All outside visible stitching: 10 to 14 stitches per inch.
	(2) Overedging or prehemming: 6 to 10 stitches per inch.
e. 401 or 101 stitch type	(1) Loops of the chain finish on the inside of the garment.
f. Overedging	(1) Knife set to trim the loose threads and raveled edge without cutting the material.
g. Seams and seam allowances	(1) Maintained with seam uniformly sewn so that no raw edges, runoffs, twists, pleats, puckers, gathers, other distortions, or open seams result.
h. Raw edges	(1) Raw edges on the outside of the garment are not allowed.
	(2) Raw edges on the inside along the double needle seams are not allowed.
	(3) Raw edges greater than ⅛ inch on the inside of the garment are not allowed.
	(4) Raw edges may be turned-in, turned-under, or serged to prevent raveling.
i. Beginning and ending of seams	(1) Start and finish evenly.
j. Joined components	(1) Lie smooth and flat.

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TABLE XIII. Construction details – Continued.

Construction Feature	Requirement
k. Ends of seams and stitching – 301 stitch type	(1) If not caught in another line of stitching: backstitched not less than ¼ inch.
	(2) If the end of a continuous line of stitching (except label): overlapped not less than ½ inch.
	(3) If label stitching: overlapped not less than 3 stitches.
l. Skipped stitches (301 or 401)	(1) Repaired by using 301 stitch type.
m. Edge, raise and topstitching	(1) Edge and raise stitching: positioned 1/16 inch from the edge.
	(2) Topstitching: positioned ¼ inch from the edge.
3. Fusible interfacings	(1) Applied in accordance with the supplier's recommendation for time, temperature, and pressure.
	(2) Centered on the basic material component part according to the pattern.
4. Buttons	(1) Securely fastened to the garment by locking stitches.
	(2) Minimum of 16 stitches per button.
	(3) "C" stitch pattern.
5. Pocket drainage eyelets	(1) Minimum of 19 stitches each.
	(2) Purling on the outside.
	(3) Material edges shall be securely caught in the eyelet stitching.
6. Buttonholes	(1) Positioned in accordance with marks on patterns.
	(2) Eyelet end, tapered bar, with ⅞-inch cut opening and 52 to 56 stitches per inch.
	(3) Buttonhole purling positioned between the fly and front waist and on the outside of the trousers.
	(4) Hip and cargo pocket buttonhole purling visible on the outside of the flap tab.
7. Bartacks	(1) Size and position in accordance with specific construction features (waistband elastic - floating, belt loops, fly closure, side pockets, hip pockets, cargo pockets, elasticized opening of cargo pocket [standard method], and crotch gusset) and as shown in figure 1 through figure 10.
8A. Waistband - standard	(1) Grown on.
	(2) Elastic enclosed.
	(3) Partially elasticized waistband providing 4 inches of waist expansion (elasticized from front pocket bearer, across outseam, and onto back of trousers).
8B. Waistband - alternate	(1) Sewn on.
	(2) Elastic enclosed.
	(3) Partially elasticized waistband providing 4 inches of waist expansion (elasticized from front pocket bearer, across outseam, and onto back of trousers).

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TABLE XIII. Construction details – Continued.

Construction Feature	Requirement
9A. Waistband elastic - floating	(1) Attached to waistband with 1- to 1⅛-inch vertical bartack at each end of the waistband elastic, thread B (see table V), 52 to 56 stiches per bartack.
	(2) Meets the elastic force vs. elongation specified in 3.5 and finished dimensions (relaxed and stretched) specified in 3.9.
9B. Waistband elastic - metered	(1) Elastic metered its entire length with the elastic inserted within the seam construction along opposite sides of the waistband per pattern placement.
	(2) Four stitch lines with correct stitch tension and balance without any impediment that could cause broken stitches when waistband is extended.
	(3) Meets the elastic force vs. elongation requirements specified in 3.5 and finished dimensions (relaxed and stretched) specified in 3.9.
10. Center front pleats	(1) Positioned in accordance with marks on patterns.
	(2) Fold of the pleat faces the outseam.
	(3) Stitched for 3±¼ inches from top of trousers.
11. Belt loops	(1) Seven belt loops located as follows: one at center back seam, one on each outseam, one on each front at the front pleat, and one on each back evenly spaced between the center back seam and outseam. Tolerance: ±¼ inch.
	(2) 1±⅛ inch wide.
	(3) Vertically aligned and parallel to each other.
	(4) ¾±1/16-inch horizontal bartacks centered across the top and bottom of each belt loop, thread B, 35 to 40 stitches per bartack.
12. Fly closure	(1) Standard (with crotch extension piece) and alternate (without crotch extension piece) construction in accordance with figure 5.
	(2) Front cut edge of buttonholes positioned ½±1/16 inch from front folded edge of the trouser.
	(3) ⅝±1/16-inch horizontal bartack at bottom of fly, superimposed on the base of the "J" stitching, thread B (see table V), 35 to 40 stitches per bartack.
	(4) ¾±1/16-inch vertical bartack positioned ¾±1/16 inch above top end of front crotch seam and ⅛ inch from outside edge of right fly. Bartack is not to be visible on outside of trouser. Thread B (see table V), 35 to 40 stitches per bartack.
13. Side pockets	(1) Positioned according to marks on patterns.
	(2) Pocket opening at least six inches.
	(3) End of stitching: ½-inch chain or secured with second row of stitching.
	(4) ⅜±1/16-inch bartacks at the top (horizontal bartack) and bottom (angled) of the side pocket openings, thread A (see table V), 20 to 27 stitches per bartack.

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TABLE XIII. Construction details – Continued.

Construction Feature	Requirement
14. Hip pockets	(1) Construction in accordance with figure 9 and figure 10.
	(2) Positioned according to marks on patterns.
	(3) Width of welt: ¼+1/16 -0 inch.
	(4) Sides and bottom edge of flap topstitched ¼ inch from edge.
	(5) Flaps set ½+⅛, -0 inch above top of pocket when measured from top of finished pocket to pocket flap setting seam. (Note: Positioning mark on pattern indicates required finished location of pocket flap).
	(6) End of stitching: ½-inch chain or secured with second row of stitching.
	(7) ⅜±1/16-inch vertical bartack at each end of hip pocket welt openings, thread A (see table V), 20 to 27 stitches per bartack.
15. Cargo pockets	(1) Construction in accordance with figure 6 through figure 8.
	(2) Positioned in accordance with marks on patterns.
	(3) Pleats face towards the back of the trouser.
	(4) Elasticized cargo pocket edge in accordance with item 16 below.
	(5) Sides and bottom edge of flap topstitched ¼ inch from the edge.
	(6) Flap finishes ½+⅛ inch above top of pocket when measured from top of finished pocket to pocket flap setting seam. (Note: Positioning mark on pattern indicates required finished location of pocket flap.)
	(7) ⅜±1/16-inch horizontal bartack at each top corner of cargo pocket flap; thread B (see table V), 20 to 27 stitches per bartack.
	(8) ⅜±1/16-inch vertical bartack at the lower front corner of the cargo pocket flap through pocket and trouser, thread B (see table V), 20 to 27 stitches per bartack.
	(9) ⅜±1/16-inch vertical barback at the top and bottom corners of the cargo pocket, thread B (see table V), 20 to 27 stitches per bartack.
	(10) ⅝±1/16-horizontal barbacks across pleat placements below lower edge of elastic on top of cargo pockets, thread B (see table V), 35 to 40 stitches per bartack.
	(11) Button closure positioned at the rear corner of flap. Positioned in accordance with marks on patterns.

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TABLE XIII. Construction details – Continued.

Construction Feature	Requirement
16. Elasticized opening of cargo pocket	(1a) Standard method – elastic encased: Each end of 1½±1/16 inch wide elastic attached to inside of pocket with a 1 to 1⅛-inch vertical bartack, then facing is attached to pocket forming casing. Thread B (see table V), 52 to 56 stitches per bartack.
	(1b) Alternate method – elastic set on: (Note: Cargo pocket facing pattern part is not used in this method). 1½±1/16-inch wide elastic attached to the pocket with two rows of double needle stitching at top and bottom edges of the elastic. No portion of the elastic is visible on the outside of the pocket. Elastic is visible from inside the pocket.
	(2) Elasticized cargo pocket edge extends when the hand is inserted into the pocket yet does not distort the trouser leg when lying in the gathered/relaxed closed position. The elastic retreats to its relaxed opening position when hand is removed.
	(3) Relaxed pocket opening measures 6¾ to 7½ inches.
	(4) The cargo pocket opening retains its elongation/recovery characteristics for the life of the garment.
17. Crotch gusset	(1) Set on the intersection of the inseam for each leg and center back and center front seams.
	(2) ⅝±1/16-inch vertical bartack where front crotch seam joins crotch gusset (see figure 4), thread A (see table V), 35 to 40 stitches per inch.
18. Reinforcement patches	(1) Set on in accordance to the pattern markings.

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TABLE XIV. Seam types.

Seam Placement	Seam Type	Gage	Stitch Type
1. Inseams	LSc-2	3/16 to 9/32-inch gage	301or 401
2. Standard outseams, back rise and front rise seams	LSc-2	3/16 to 9/32-inch gage	301or 401
3. Alternate outseams, back rise and front rise seams	SSa-2 and LSq-3	Overedge 3/16 to 9/32-inch gage, then turn and topstitch 3/16 to 1/4 inch	516 and 301 or 401
4. Standard waistband (grown-on)	LSk-4 (modified)	Waistband 1½ inches with 2 rows of double needle stitching	301 or 401
5. Alternate waistband (sewn-on)	LSbc-4 (modified)	Waistband 1½ inches with 2 rows of stitching securing the elastic	301 or 401
6. Topstitching for fly	OSf-1	1/8 to 1/4 inch from the edge	301
7. Closing of side hanging and hip pockets	SSe-2 and OSf-1 or SSa-2	3/16 to 1/4 inch from edges	301 or 515, 516, or 519
8. Hip pocket opening	SSbe-2	Welt exterior 1/4 inch, interior 1±1/8 inch	301
9. Attachment of side and hip pocket bearers and facings	LSd-1	1/16 to 1/8 inch from the finished edge	301
10. Standard elasticized cargo pocket opening	EFr-2	One row 1/16 to 1/8 inch from top pocket edge and the bottom row 1/16 to 1/8 inch from folded bottom edge of facing	301 or 401
11. Alternate elasticized cargo pocket opening	SSt-4	(a) Two rows of double needle stitching, 1/4-inch gauge, at top and bottom edge of elastic (b) Outer row of stitching shall be 1/16 to 1/8 of an inch from each edge of elastic	301 or 401
12. Pocket flap attachment	LSbj-2 (modified)	Two rows of double needle stitching 3/16 to 1/4 inch apart	301
13. Belt loop construction	EFh-1	5/16 to 3/8 inch from edges	406
14. Attachment of seat and knee patches	LSd-2	Edgestitched with two rows 3/16 to 1/4 inch apart	301
15. Label attachment	LSbj-1	1/8 to 3/16 inch from edges	301
16. Trousers leg hems	EFb-1	1/16 to 1/8 inch from edge	301

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3.9 Finished dimensions. Finished dimensions are provided below for the finished trousers (either made from insect repellent treated basic material or after application of an insect repellent treatment in the garment form). The finished alpha sizing trousers shall conform to the dimensions of table XV, and the finished numeric sizing trousers shall conform to the dimensions of table XVI, as defined in table XVII, when tested as specified in 4.3.5. All measurements shall be taken with the garment buttoned and laid flat on a table. Dimensions and tolerances are expressed in inches.

TABLE XV. Alpha sizing finished trouser dimensions.

Letter	Measurement	Size/ Length	X-Small	Small	Medium	Large	X-Large	XX-Large	Tolerance
A	**Half waist – relaxed**	All lengths	12	14	16	18	20	22	-¼, +½
B	**Half waist – stretched**	All lengths	14	16	18	20	22	24	-0, +½
C	**Half seat**	All lengths	19½	21½	23½	25½	27½	29½	-¼, +½
D	**Bottom**	All lengths	17	17	17¾	17¾	18½	18½	±½
E	**Front rise**	X-Short	10	10½	11	11½	-	-	
		Short	10½	11	11½	12	12½	-	
		Regular	11	11½	12	12½	13	13½	
		Long	11½	12	12½	13	13½	14	
		X-Long	-	12½	13	13½	14	14½	
		XX-Long	-	-	13½	-	-	15	
F	**Back rise**	X-Short	15½	15⅞	16¼	16⅝	-	-	
		Short	15⅞	16¼	16⅝	17	17⅜	-	
		Regular	16¼	16⅝	17	17⅜	17¾	18⅛	
		Long	16⅝	17	17⅜	17¾	18⅛	18½	
		X-Long	-	17⅜	17¾	18⅛	18½	18⅞	
		XX-Long	-	-	18⅛	-	-	19¼	
G	**Inseam**	X-Short	28¾	28¾	28¾	28¾	-	-	±¾
		Short	30¼	30¼	30¼	30¼	30¼	-	
		Regular	32¾	32¾	32¾	32¾	32¾	32¾	
		Long	34¾	34¾	34¾	34¾	34¾	34¾	
		X-Long	-	36¾	36¾	36¾	36¾	36¾	
		XX-Long	-	-	38¾	-	-	38¾	
H	**Outseam**	X-Short	37¾	38¼	38¾	39¼	-	-	±¾
		Short	40¼	40¾	41¼	41¾	42¼	-	
		Regular	42¾	43¼	43¾	44¼	44¾	45¼	
		Long	45¼	45¾	46¼	46¾	47¼	47¾	
		X-Long	-	48¼	48¾	49¼	49¾	50¼	
		XX-Long	-	-	51¼	-	-	52¾	

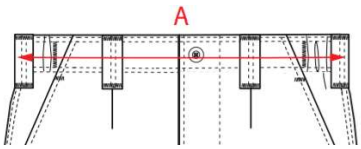
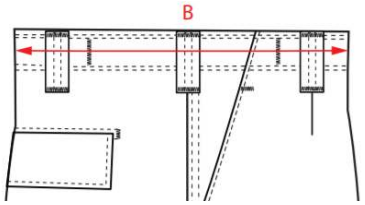
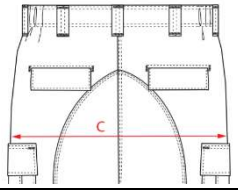
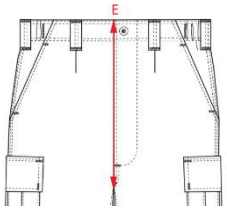
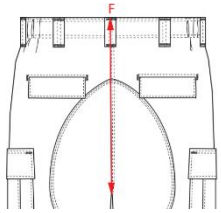
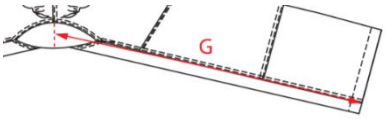
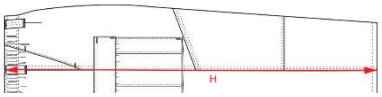
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TABLE XVI. Numeric sizing finished trouser dimensions.

Letter	Measurement	Size/ Length	24	28	32	Tolerance
A	**Half waist – relaxed**	All lengths	12	13½	15	-¼, +½
B	**Half waist – stretched**	All lengths	14	15½	17	-0, +½
C	**Half seat**	All lengths	19¾	21¾	22¾	±¾
D	**Bottom**	All lengths	16¼	16¼	17	±½
E	**Front rise**	X-Short	11¼	11⅝	-	±¼
		Short	11¾	12⅛	12½	
		Regular	-	12⅝	-	
F	**Back rise**	X-Short	15⅜	15¾	-	±¼
		Short	15⅞	16¼	16⅝	
		Regular	-	16¾	-	
G	**Inseam**	X-Short	30	30	30	±¾
		Short	32	32	32	
		Regular	-	34	-	
H	**Outseam**	X-Short	39⅛	39½	39⅞	±¾
		Short	41⅝	42	42⅜	
		Regular	-	44½	-	

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TABLE XVII. Methods of measure for finished trousers dimensions (alpha and numeric).

Letter	Measurement	Method	Measurement Depiction
A	**Half waist – relaxed**	With elastic inserts relaxed, along center of waistband from outside folded edge to folded edge.	
B	**Half waist – stretched**	With ½ waist rotated so that elastic is in the approximate center and waist stretched until fully extended without excessive force, horizontally across center of waistband from outside folded edge to folded edge.	
C	**Half seat**	Horizontally across from folded edge to folded edge 1 inch above the cargo pocket flaps.	
D	**Bottom**	Horizontally across bottom of leg from folded edge to folded edge. Multiply by two.	
E	**Front rise**	From top of waistband at center front vertically down to top of the gusset at the joining seam.	
F	**Back rise**	From top of waistband at center back vertically down to top of the gusset at the joining seam.	
G	**Inseam**	From center of gusset to bottom edge of trouser leg.	
H	**Outseam**	From top edge of waist to bottom of leg along outseam.	

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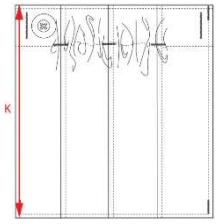
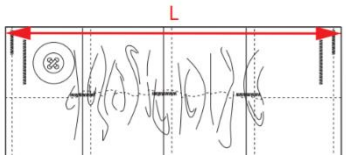
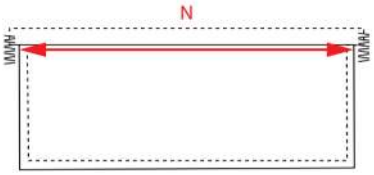
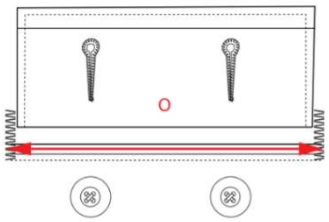
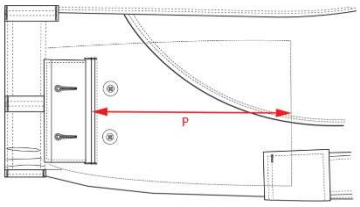
3.9.1 Finished trouser pocket and flap dimensions. The pockets and flaps of finished trousers shall conform to the dimensions of table XVIII, as defined in table XIX, when tested as specified in 4.3.4.2.

TABLE XVIII. Finished trouser pocket and flap dimensions (alpha and numeric).

Letter	Measurement	Length	Size	Dimension	Tolerance
I	**Cargo pocket flap, length**	All lengths	All sizes	2⅜	±¼ inch
J	**Cargo pocket flap, width**	All lengths	All sizes	8⅛	±¼ inch
K	**Cargo pocket, length**				
	Alpha	X-Short	X-Small	8¼	±¼ inch
			Small		
			Medium	9	±¼ inch
			Large		
		Short	X-Small	9	±¼ inch
			Small		
			Medium	10	±¼ inch
			Large		
			X-Large		
		Regular	All sizes	10	±¼ inch
		Long			
		X-Long			
		XX-Long			
	Numeric	X-Short	24, 28	8¼	±¼ inch
		Short	24, 28	9	±¼ inch
			32	10	±¼ inch
		Regular	28	10	±¼ inch
L	**Cargo pocket, width (relaxed)**	All lengths	All sizes	6¾	±¼ inch
M	**Hip pocket flap, length**	All lengths	All sizes	2¼	±¼ inch
N	**Hip pocket flap, width**	All lengths	All sizes	6	±¼ inch
O	**Hip pocket, opening**	All lengths	All sizes	5½	±¼ inch
P	**Hip pocket, depth**	All lengths	All sizes	6¼	±¼ inch

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TABLE XIX. Methods of measure for finished trouser pocket and flap dimensions (alpha and numeric).

Letter	Measurement	Method	Measurement Depiction
I	**Cargo pocket flap, length**	From the left side, top edge vertically down to the left side, bottom edge.	
J	**Cargo pocket flap, width**	From the left side, top edge horizontally across to the right side, top edge.	
K	**Cargo pocket, length**	From the left side, top edge of the cargo pocket vertically down to the left side, bottom edge.	
L	**Cargo pocket, width (relaxed)**	From the left side, top edge of the cargo pocket horizontally across to the right side, top edge with the elastic relaxed.	
M	**Hip pocket flap, length**	From the left side, top edge vertically down to the left side, bottom edge.	
N	**Hip pocket flap, width**	From the left side, top edge, horizontally across to the right side, top edge.	
O	**Hip pocket, opening**	From the left bartack horizontally across to the right bartack.	
P	**Hip pocket, depth**	From the top of the welt vertically down to the bottom of the pocket	

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3.10 Workmanship. The treated trousers shall conform to the quality of product established by this document, be uniform, and be free from defects specified in 4.3.4.2 and those defects that adversely affect form, fit, or function. Defects shall not exceed the established Acceptance Quality Limits (AQLs) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

3.10.1 Disposal of class 1 and 2. For class 1 and 2, all scraps, garment seconds, irregulars, extra material, and extra garments shall be disposed of as specified in the contract (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract (see 6.2), the first article, in accordance with 3.1.1, shall be examined in accordance with 4.3.1 through 4.3.5 for compliance with the appearance, color, finish, design, configuration, workmanship, dimensional, and testing requirements as specified in this document.

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), first article samples shall be provided. The sample size and the sample unit shall be as specified in the contract (see 6.2). The lot size shall be expressed in units of trousers. The presence of excessive defects or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests as specified in 4.3.1 through 4.3.5. The Government reserves the right to inspect and test all components, materials, and end items to determine conformance to stated requirements.

4.3.1 Materials and components (see 3.4).

4.3.1.1 Sampling and acceptance criteria for verification of materials and components. Unless otherwise specified in referenced documents or procurement documents (see 6.2), material and component sampling shall be in accordance with table XX. The unit for expressing lot size and sample unit shall be in accordance with the applicable reference documents. The component lot shall be unacceptable if one or more sample units fail to meet any test requirements specified in this document.

TABLE XX. Materials and component sampling.

Lot Size	Sample Size (# of Sample Units)
800 or less	2
801 through 22,000	3
22,001 and over	5

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4.3.1.2 Materials and components examinations and tests. In accordance with 3.4, the materials and components shall be examined and tested in accordance with the specified examinations and tests referenced in table XXI unless otherwise excluded, amended, or modified in this document or applicable procurement documents (see 6.2).

TABLE XXI. Material and component verification.

Material/Component	Requirement	Verification
Treated basic material 1/	3.4.1	MIL-DTL-32679
Reinforcement material	3.4.2	MIL-DTL-32439
Interfacing	3.4.3	MIL-C-44296
Thread	3.4.4	Per specific thread specification listed in table V
Elastic webbing	3.4.5	MIL-W-5664 and visual
Buttons	3.4.6	A-A-59994
Button shade	3.4.6	AATCC EP9, option A
Labels	3.4.7	Per requirements specified in table VIII

FOOTNOTE:

1/ When the alternate untreated basic material is used and the insect repellent is applied in garment form, the verification of the material shall be conducted in accordance with 4.3.2.

4.3.2 End item physical requirements.

4.3.2.1 Sampling and acceptance criteria for verification of end item physical requirements. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed conforming to ASQ/ANSI Z1.4; the lot size shall be expressed in units of finished trousers, and the sample unit shall be six trousers, and the presence of excessive defects or failure of any testing as specified shall be cause for rejection of the lot.

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4.3.2.2 End item physical requirements examination and tests. The finished end items shall be examined and tested in accordance with the specified examinations and tests referenced in table XXII unless otherwise excluded, amended, or modified in this document or applicable procurement documents (see 6.2). Garment physical tests apply to finished trousers made from treated basic material or to finished trousers made from the alternate basic material that have been treated with insect protection in the garment form.

TABLE XXII. End item physical requirements verification.

Characteristic	Requirement	Verification
Alternate basic material (after garment has been treated with insect repellent)		
Class 1 trouser	3.5	MIL-DTL-32679, type II, class 1
Class 2 trouser	3.5	MIL-DTL-32679, type II, class 2
Garment physical tests		
Dry out time (Medium-Regular size)	3.5	See 4.3.2.3
Trouser weight (Medium-Regular size)	3.5	See 4.3.2.4
Button pull	3.5	See 4.3.2.5
Elastic force vs. elongation		
Cargo pocket	3.5	See 4.3.2.6 and 4.3.2.6.1
Waistband	3.5	See 4.3.2.6 and 4.3.2.6.2
Toxicity	3.5.1	See 4.3.2.7

4.3.2.3 Dry out time. The finished trousers shall be tested for dry out time. The trousers shall be washed and dried one cycle before conditioning and measurements. After laundering, the trousers shall be conditioned at 90 °F and 80 percent relative humidity (RH) for a minimum of 24 hours and then weighed. This is the dry mass. After recording the dry mass, the trousers shall be thoroughly soaked in water at 80 °F for 4 minutes in an AATCC-approved washer and then run through the spin cycle. After the spin cycle, the trousers shall be removed and weighed. This is the spun mass and the weight shall be recorded. The trousers shall be placed in an environmental test chamber with temperature and humidity set at 90 °F and 80 percent RH. The weight of the trousers shall be continuously recorded. The trousers dry out time is the time elapsed when there is 4 ounces of water remaining in the trousers.

4.3.2.4 Trouser weight. The finished trousers shall be conditioned for a minimum of 24 hours at standard laboratory conditions (i.e., 70 °F and 65 percent RH) and then weighed.

4.3.2.5 Button pull test. The buttons attached to the trousers shall meet the requirements stated in 3.6 when tested in accordance with ASTM D5034, except that the button shall be slid on top of 1-inch wide grips separated at least ⅜ inch so that the button loosely rests evenly on top of grips. Only manually adjusted grips shall be allowed, not pneumatic. The button attachment stitching shall be placed in the middle of the bottom grips and adjusted an additional ⅜-inch down so that the grips grab the fabric just below the stitching, not on the stitching. The tensile machine shall be run per the test method, and the force required to pull button from sewn garment shall be recorded.

4.3.2.6 Force vs elongation of elasticized features. Cargo pocket and waistband samples shall be samples from trousers that are constructed to duplicate the configuration on the finished garment where the elastic webbing is set on or within the waistband. The cargo pocket length (relaxed) shall be the finished opening length of 6 inches. The waistband shall consist of one side of the elasticized waistband, which is approximately 4 inches in length (relaxed). Both cargo pocket and waistband samples shall be laundered five times as per AATCC TM135, 3, V, iii, and stitching that sets the elastic shall stay secured through the washings. Tensile properties shall be evaluated by conducting tensile tests utilizing equipment that conforms to ASTM D76/D76M. Samples shall be conditioned in a 70 °F, 65 percent RH laboratory for a minimum of 24 hours. The specimens shall be clamped in pneumatic jaws in a relaxed state (no force on read out) when the jaws are spaced 4.0 inches apart.

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4.3.2.6.1 Cargo pocket. The cargo pocket sample shall be centered between the jaws so that the set elastic webbing is clamped into each jaw. A constant rate of extension (CRE) with a loading rate of 2.0 inches per minute shall be used. The crosshead shall be moved 1.5 inches for a total of 5.5 inches. The resulting force shall be recorded at a rate of 10 points per second. One cycle shall consist of this elongation as well as the return to the initial 4.0-inch separation position. The cycle shall be repeated 100 times for each sample. The graph shall be evaluated, and the sample shall meet the requirement when no peak is greater than 3.5 pounds or less than 2.5 pounds.

4.3.2.6.2 Waistband. A CRE with a loading rate of 2.0 inches per minute shall be used. The crosshead shall be moved 1.5 inches for a total of 5.5 inches. The resulting force shall be recorded at a rate of 10 points per second. The graph shall be evaluated to record the force separately at ¾-inch and 1½-inch extension. The sample shall meet the requirements when the ¾-inch extension is not greater than 1.5 pounds or less than 0.5 pound, and the 1½-inch extension is not greater than 4.0 pounds or less than 1.5 pounds. The sample shall demonstrate acceptable performance at both extensions to pass the requirement.

4.3.2.7 Toxicity test. When required (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (Hayes' Principles and Methods of Toxicology). If toxicity requirements (see 3.5.1) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

4.3.3 In-process examination. Visual examinations shall be made at any point or during any phase of the manufacturing process to determine whether material or construction details, which cannot be examined in the finished product, are in accordance with the requirements specified herein. Materials that contain defects and damages, as defined in the visual examination of 4.3.4, shall be removed from production. Any in-process nonconformance remaining in the finished material shall be classified as a defect in accordance with the visual examination of 4.3.4.

4.3.4 Visual inspection of end items.

4.3.4.1 Sampling and acceptance criteria for visual inspection of end items. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed conforming to ASQ/ANSI Z1.4; the lot size shall be expressed in units of trousers, and the sample unit shall be one pair of trousers; and the presence of excessive defects or failure of any testing as specified shall be cause for rejection of the lot.

4.3.4.2 Visual inspection. The trousers shall be examined for defects in color, shade, design, materials, construction, workmanship, and marking. Defects shall be classified in accordance with the list of defects specified in table XXIII. Table XXIV provides guidance regarding methods of measurement to determine evenness between garment parts, which include, but are not limited to, inseams, pocket alignment, etc. The inspection conditions and AQLs shall be as specified (see 6.2).

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TABLE XXIII. Defects.

Part	Defects	Classification	
		Major	Minor
1. Material	a. Defects are as specified in ASTM D3990 and including, but are not limited to, holes, slubs, cuts, tears, abrasion marks, smash, burns, drill holes, runs, needle chew, missing or broken yarns, visible mends, knots, thin place, dye streaks, or misweaves affecting appearance or serviceability.	101	
2. Camouflage pattern (basic and reinforcement material)	a. Camouflage pattern design not equal to standard sample or drawing, incorrect color in any part of the pattern.	102	
	b. Camouflage pattern repeat not equal to the standard sample or drawing.	103	
	c. Skitteriness (mottles, uneven color) of pattern exceeds that shown by the standard sample.	104	
	d. Grinning (off register, gap where ground shade shows through) of pattern exceeds that shown by the standard sample.	105	
	e. Haloing or trapping (overlapping of colors) of pattern exceeds that shown by the standard sample.	106	
3. Material and components assembly	a. Material or component part not as specified or omitted.	107	
	b. Distorted, full, tight, or twisted; any part of the trousers caught in any unrelated stitching.	108	
	c. The edge of any component part required to be forced out having folds of more than ⅛ inch.		201
	d. Fullness creating unwanted permanent fold, pleat, or crease.	109	
	e. Any required operation or assembly not as specified, omitted, or improperly performed.	110	
	f. Component part not cut on warp or as specified.	111	
	g. Bubbling, delaminating, wrinkling, puckering.		202
4. Shade	a. Shade variation within a part or between parts. 1/		203
	b. Component part color not as specified.		204
5. Trouser evenness 2/	a. Waistband uneven by more than ¼ inch when buttoned.		205
	b. Inseam or outseam lengths vary by more than ½ inch from leg to leg.	112	
	c. Bottom openings vary by more than ½ inch in half width from leg to leg.		206
	d. Evenness of length between inseam and outseam on the same leg varies by more than ½ inch.		207

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TABLE XXIII. Defects – Continued.

Part	Defects	Classification	
		Major	Minor
6. Stitching and seams	a. Seam, seam allowance, or stitch type not as specified.		208
	b. Seams irregular, puckered, distorted, pleated, wavy, full, or twisted.		209
	c. Open seam, raw edge on outside, raw edge on inside along double needle seam, raw edge greater than ⅛ inch on inside of garment.		210
	d. Raw edge:		
	(1) On outside.	113	
	(2) On inside:		
	(a) Up to ½ inch.		211
	(b) ½ inch or greater.	114	
	e. Loose or tight stitch tension, broken or missing thread or stitch.		212
	f. Broken stitching not repaired as specified.		213
	g. End of seam not secured as specified.	115	
	h. Double needle intersecting seams staggered by more than ¼ inch.		214
	i. Edge or raised stitching sewn too close to edge resulting in damage to cloth, needle chew, visible mend.		215
	j. Run off for edge and raise stitching greater than ⅛ inch.		216
7. Bartacks	a. Bartack missing or not secure.		217
	b. Left fly crotch bartack or J stitching bartack not positioned as specified.		218
	c. Bartack other than left fly crotch bartack or J stitching bartack not positioned as specified.		219
	d. Bartack not required length or width.		220
	e. Number of stitches not as specified or broken stitches.		221
8. Fly lining	a. Trousers fly lining exposed more than 1/16 inch.		222
9. Pleats	a. Trouser pleat omitted or reversed (i.e., not facing outseam).		223
	b. Front panel pleat stitched down less than 2¾ or more than 3¼ inches from top of trousers.		224
10. Elastic features	a. Waist elastic:		
	(1) Omitted, twisted, not positioned as specified, not attached as specified.	116	
	(2) Floating method: not caught in bartack.	117	
	b. Cargo pocket elastic:		
	(1) Omitted, twisted, not positioned as specified, not attached as specified.	118	
	(2) Setting distorts trouser leg.	119	
	(3) Standard method: not caught in bartack.	120	
	(4) Does not expand/retract to relaxed condition as specified.		225

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TABLE XXIII. Defects – Continued.

Part	Defects	Classification	
		Major	Minor
11. Belt loops	a. Omitted, not securely stitched, or not positioned as specified.	121	
	b. Width less than ⅞ or more than 1⅛ inch.		226
	c. Opening less than 2¾ or more than 3¼ inches long.		227
	d. Belt loop set at an angle (more than 3/16 inch off perpendicular).		228
12. Hems	a. Hem at leg bottom twisted, wavy, omitted, or not as specified.		229
	b. Hem width of leg bottom less than ½ or more than ¾ inch. (NOTE: Hem measurement taken from top to bottom fold.)		230
13. Buttons and buttonholes	a. Any button and buttonhole out of alignment causing bulge, twist, or distortion when buttoned.	122	
	b. Cut opening of pocket flap buttonhole caught in topstitching.	123	
	c. Buttonhole omitted or added, type not as specified, location or direction not as specified, cut length not as specified.	124	
	d. Buttonhole not clean cut or not securely caught in fabric.	125	
	e. Buttonhole opening greater than ⅞ inch.		231
	f. Gimp omitted from buttonhole or not as specified.	126	
	g. Button missing, broken, defective, or insecurely sewn.		232
	h. Stitching not locked at end of cycle (determined by tugging loose end of thread).		233
	i. Purling not on side specified.		234
	j. Button not attached with “C” stitch pattern.	127	
14. Pockets and pocket flaps	a. Pocket companions not uniform in size or shape.		235
	b. Pocket or flap not stitched as specified, not positioned as specified, incorrect size.		236
	c. Pocket or pocket flap twisted, curled, or puckered.		237
	d. Pocket flap not completely covering pocket opening.		238
	e. Pocket flaps not centered over pocket opening. Flap width extends past finished pocket edge by more than ⅛ inch on either side.		239
	f. Cargo pocket bellows exposed beyond edge of pocket by more than ⅛ inch.	128	
	g. Cargo or hip pockets out of horizontal alignment by more than ½ inch.	129	
	h. Any pocket or pocket flap measurement not as specified.		240
	i. Cargo pocket elasticized opening less than 6¾ or more than 7½ inches in width, forward bottom corner of flap not bartacked down.		241
	j. Cargo pocket bartack not stitched through both flap and pocket.		242
	k. Side hanging pocket opening less than 6 inches.	130	

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TABLE XXIII. Defects – Continued.

Part	Defects	Classification	
		Major	Minor
15. Eyelets	a. Omitted, added, misplaced, improper size or type, or caught in top stitching.		243
	b. Not clean cut, not securely caught in material.		244
	c. Eyelet not positioned as specified.		245
16. Reinforcement patches	a. Not attached as specified or not positioned as specified on pattern.	131	
	b. Knee patches out of horizontal alignment by more than ½ inch.		246
	c. Top of the seat patches out of alignment by more than ¼ inch at the seat seam.		247
	d. Any opening or gap in the crotch reinforcement area caused by incorrect positioning or stitching.	132	
17. Labels	a. Label omitted; incorrect type, size, color, or configuration; illegible; incorrect inscription; spelling errors; improperly attached; stitching through the text; not attached as specified; peeling.	133	
	b. Barcode not readable by scanner, human-readable interpretation omitted, or illegible.	134	
	c. Barcode not visible on folded, packaged item.		248
	d. Barcode attachment causes damage to the item.	135	
18. Cleanliness	a. Spot or stain (more than 1 inch), odor, embedded foreign matter, overall uncleanliness.	136	
	b. No more than two thread ends not trimmed to ½ inch or less on outside of garment, no more than five thread ends not trimmed to ½ inch or less on inside of garment.		249
	c. Five or more loose threads of 1 inch or more not removed throughout the garment (inside and outside combined).		250
	d. Two or more tickets not removed.		251
19. Packaging	a. Any trousers not packaged in accordance with the contract or purchase order.		252

FOOTNOTES:

1/ Parts suspected of being off shade shall be examined at a distance of 3 feet against a background of the other parts and colors of the garment. When the shade difference is discernable under these examination conditions, it shall be scored as a shaded part.

2/ Evenness can be determined by comparing measurements of companion parts or by aligning parts and measuring the difference in lengths as described in table XXIV. Evenness comparisons shall be conducted by extending the portions of the garments in parallel between two hands without stretching and then inspecting for evenness. Where evenness defects are found, the garments shall be laid flat on a measuring surface and the difference in lengths recorded.

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TABLE XXIV. Evenness verification.

Letter	Location	Description
A	Trouser inseam	Align both trouser leg inseams along the top edge, starting from the center of the crotch seam along the inseam down to the trouser hem edge. The aligned trouser legs shall be inspected for evenness of the trouser length.
B	Trouser outseam	Align both trouser leg outseams by folding trousers in half width wise following the crease of the legs and aligning the top of the waistband. The aligned trouser legs shall be inspected for evenness of the trouser length.
C	Trouser inseam and outseam uniformity	While measuring inseam, the uniformity in length between inseam and outseam shall be compared.
D	Knee patch alignment	Align both trouser leg outseams by folding trousers in half width wise following the center of each of the legs and aligning the top of the waistband in half. The aligned trouser legs shall be inspected for knee patch alignment.
E	Cargo pocket alignment	Align both trouser leg outseams by folding trousers in half width wise following the crease of the legs and aligning the top of the waistband. The aligned trouser legs shall be inspected for cargo pocket alignment.
F	Bottom opening	Align both trouser leg inseams and outseams at the bottom hem by folding trousers in half width wise following the crease of the legs and aligning the top of the waistband. The folded trouser shall be turned vertically to enable alignment of the bottom opening widths for comparison. The aligned trouser leg bottoms shall be inspected for bottom opening width.
G	Seat patch alignment	Assessed at the seat seam with the trousers laid flat.

4.3.5 Dimensional examination of end items.

4.3.5.1 Sampling and acceptance criteria for dimensional examination of end items. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed conforming to ASQ/ANSI Z1.4. The lot size shall be expressed in units of trousers, and the sample unit shall be one pair of trousers. The presence of excessive defects as specified (see 6.2) shall be cause for rejection of the lot.

4.3.5.2 Dimensional examination. The trousers shall be examined for conformance to the dimensions specified in table XV and table XVI when measured as specified in table XVII. Any measurement deviating from the dimensions and tolerances specified shall be scored as a measurement defect. The inspection conditions and the AQLs shall be as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The trousers are a component of the MCTCU for wear by military personnel of the United States Marine Corps in hot weather combat environments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classes, sizes and lengths required (see 1.2)

c. National stock numbers (NSNs).

d. When first article is required (see 3.1.1, 4.2, and 6.4).

e. When conformance inspection is required (see 3.1.2).

f. When Government approval of the label inscription is required (see 3.4.7).

g. When off-shade basic material of the same basic material type can be utilized for bagged pockets only (see table III and 3.7.3).

h. Applicable Government patterns, drawings, standard samples, and reference standards, including revisions (see 3.7).

i. When Government approval of the contractor's working files is required (see 3.7.2).

j. AQLs (see 3.10, 4.3.4.2, and 4.3.5.2).

k. Disposal of class 1 and class 2 (see 3.10.1 and 6.6.1).

l. First article sampling and acceptance criteria, including sample size and sample unit (see 4.2.1).

m. Conformance inspection criteria (see 4.3).

n. Materials and components sampling and acceptance criteria (see 4.3.1.1).

o. Materials and components examinations and tests (see 4.3.1.2).

p. End item physical requirements sampling and acceptance criteria (see 4.3.2.1).

q. End item physical requirements examination and tests (see 4.3.2.2).

r. When toxicity testing is required (see 4.3.2.7).

s. Visual examination sampling and acceptance criteria (see 4.3.4.1).

t. Visual inspections conditions (see 4.3.4.2).

u. Dimensional examination sampling and acceptance criteria (see 4.3.5.1).

v. Dimensional examination (see 4.3.5.2).

w. Packaging requirements, including insect protection labeling (see 5.1 and 6.9).

6.3 Information requests. For access to information such as specifications, drawings, and standard shade samples of cloth, go to https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx for the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Pattern/Drawing request form. Complete the request form and then submit the form. Request to use equivalent materials or components should be sent to the contracting officer for approval by the military service.

6.4 First article. When first article inspection (see 3.1.1) is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2.1. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

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6.5 Replacement document (in part). This document replaces the trousers requirements previously stated in MIL-PRF-MCTCU dated APR 2020.

6.6 Government intellectual property and trademark rights notification. The Government possesses intellectual property/trademark rights for the following United States Marine Corps patterns and logos (collectively referred to as "intellectual property"): the Eagle, Globe, and Anchor (EGA) logo, including the EGA logo as it appears embedded in the fabric pattern. The Government further has title to the invention disclosed and claimed in United States Design Patent No. D491,372 issued on 15 June 2004 for "Camouflage Pattern for Sheet Material and Uniforms" and the invention disclosed and claimed in United States Patent No. 6,805,957 issued on 19 October 2004 for "Camouflage U.S. Marine Corps Utility Uniform: Pattern, Fabric, and Design." The Government claims exclusive ownership of the above mentioned intellectual property; therefore, no entity other than the Government, or those contracted by or having obtained proper permission or licenses from the Government to do so, are permitted to produce, sell, or transfer in any manner any items (clothing or non-clothing) containing or copying, in whole or in part, the intellectual property. Doing so will be considered an infringement on the Government's intellectual property rights and will be subject to legal action.

6.6.1 Disposal of class 1 and class 2. For class 1 and class 2, all scraps, seconds, irregulars, extra material, and garments containing the aforementioned intellectual property/trademarks that are not utilized for Government contracts or a purpose authorized in writing by the Government are to be destroyed and not sold or transferred in any manner. This restriction applies to the prime contractor, as well as all subcontractors, and is to be incorporated into all agreements with subcontractors (see 6.2).

6.7 Button source. A known source of the buttons is Emsig Manufacturing Corporation (www.usbutton.com): shade M3738 for the class 1 trouser and shade M2138 for the class 2 trouser.

6.8 NSNs. For informational purposes only, NSN 8415-01-680-3232 has been assigned to the class 1 Medium-Regular trouser.

6.9 Class 1 and class 2 insect repellent treated item packaging. Unless otherwise specified (see 6.2), packaging of class 1 and class 2 trousers should be in accordance with MIL-PRF-32659.

6.10 Subject term (key word) listing.

Camouflage

Combat

Desert

Insect repellent

Jungle

Marine Corps

MCTCU

Woodland

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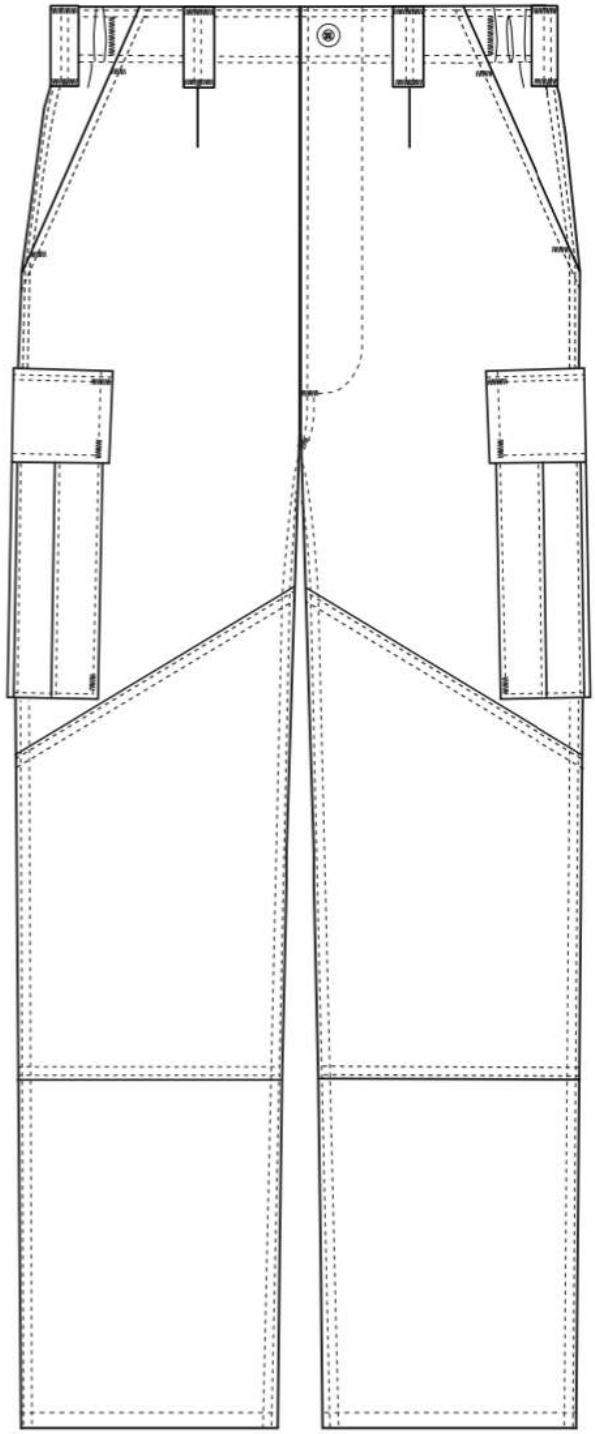


FIGURE 1. Front.

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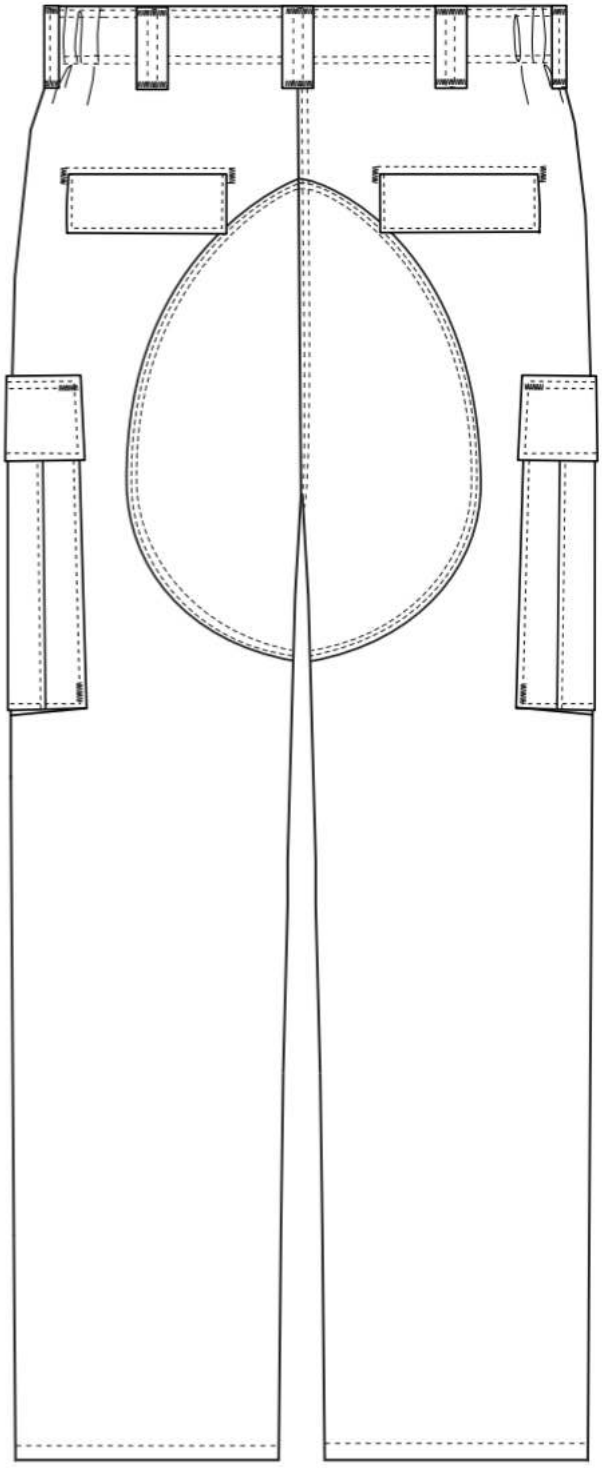


FIGURE 2. Back.

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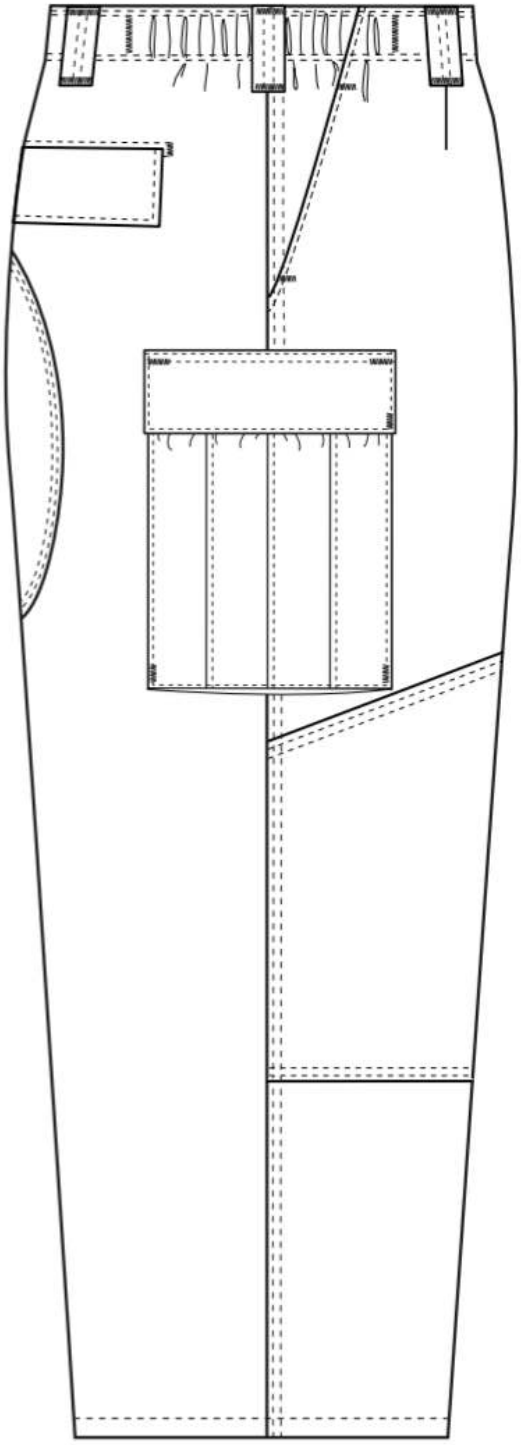


FIGURE 3. Side.

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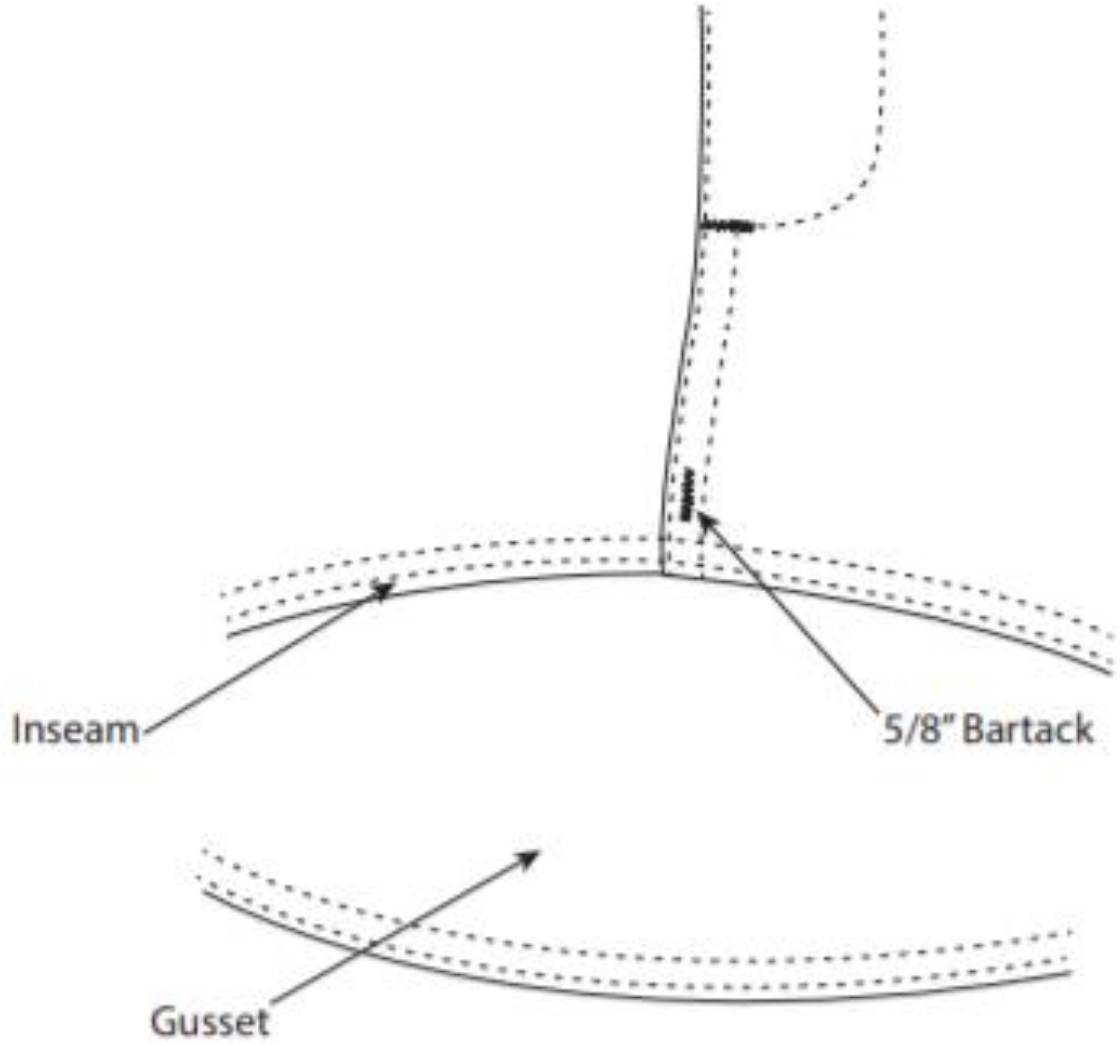
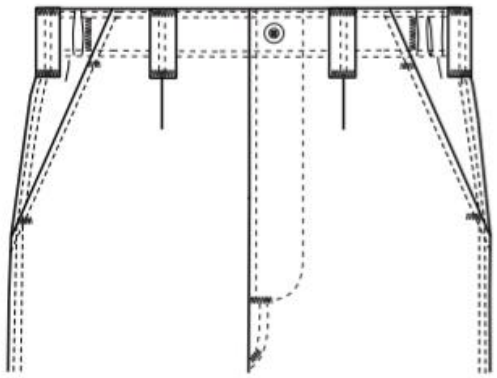
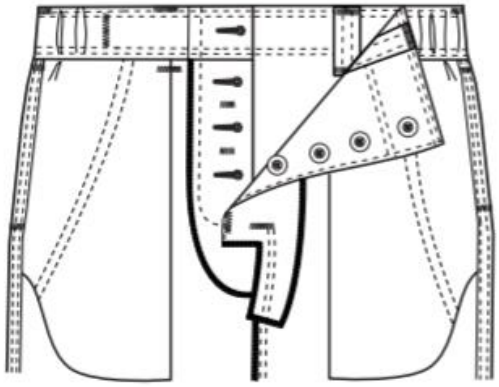


FIGURE 4. Gusset.

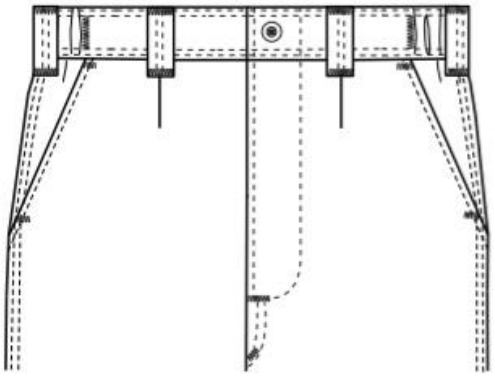
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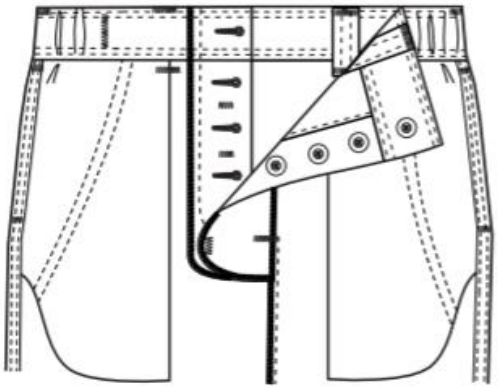
Standard Waist Construction
"Grown on"
Outside View



Standard Fly Construction
Inside View



Alternate Waist Construction
"Sewn on"
Outside View



Alternate Fly Construction
Inside View

FIGURE 5. Waist and fly construction (standard and alternate).

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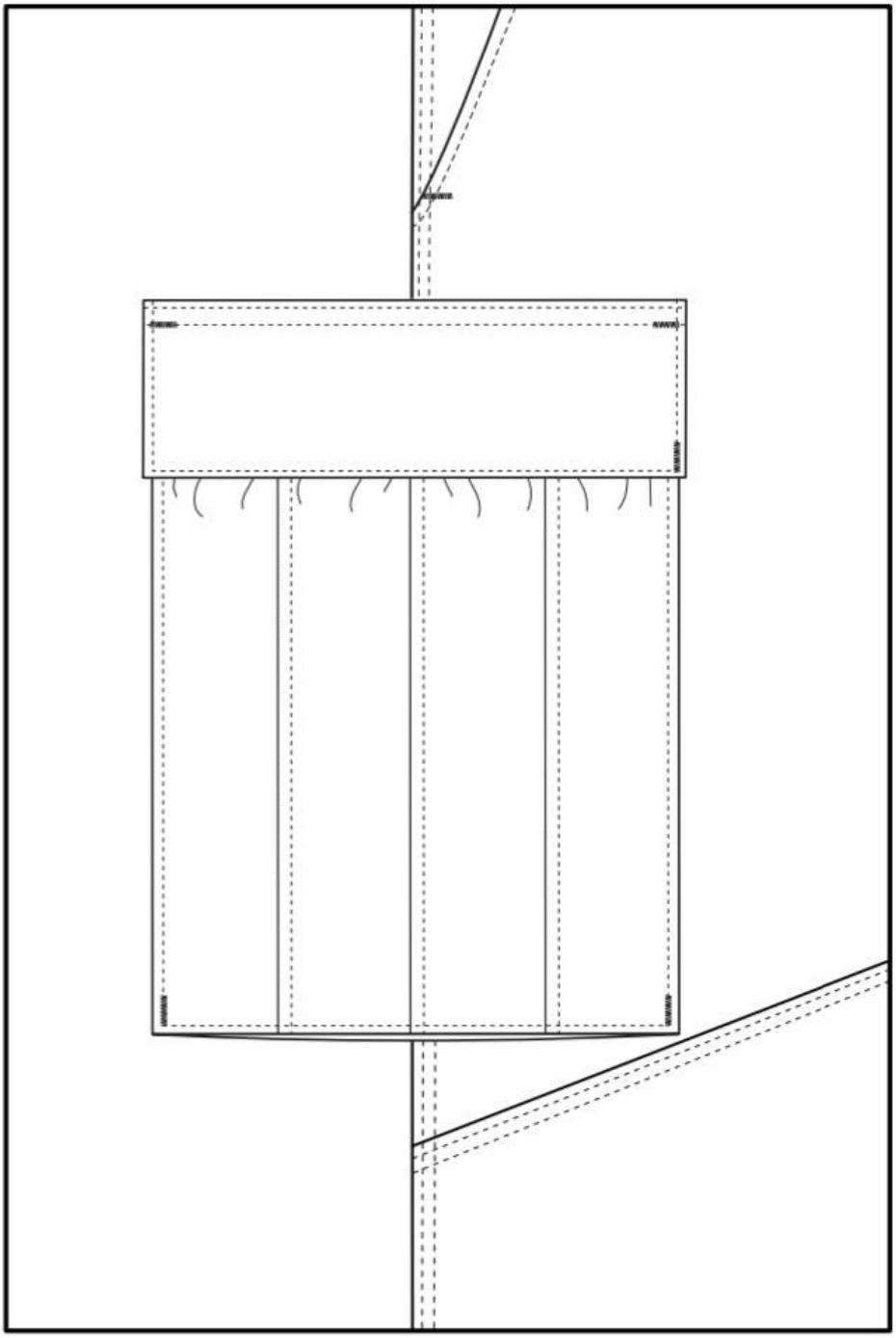


FIGURE 6. Cargo pocket.

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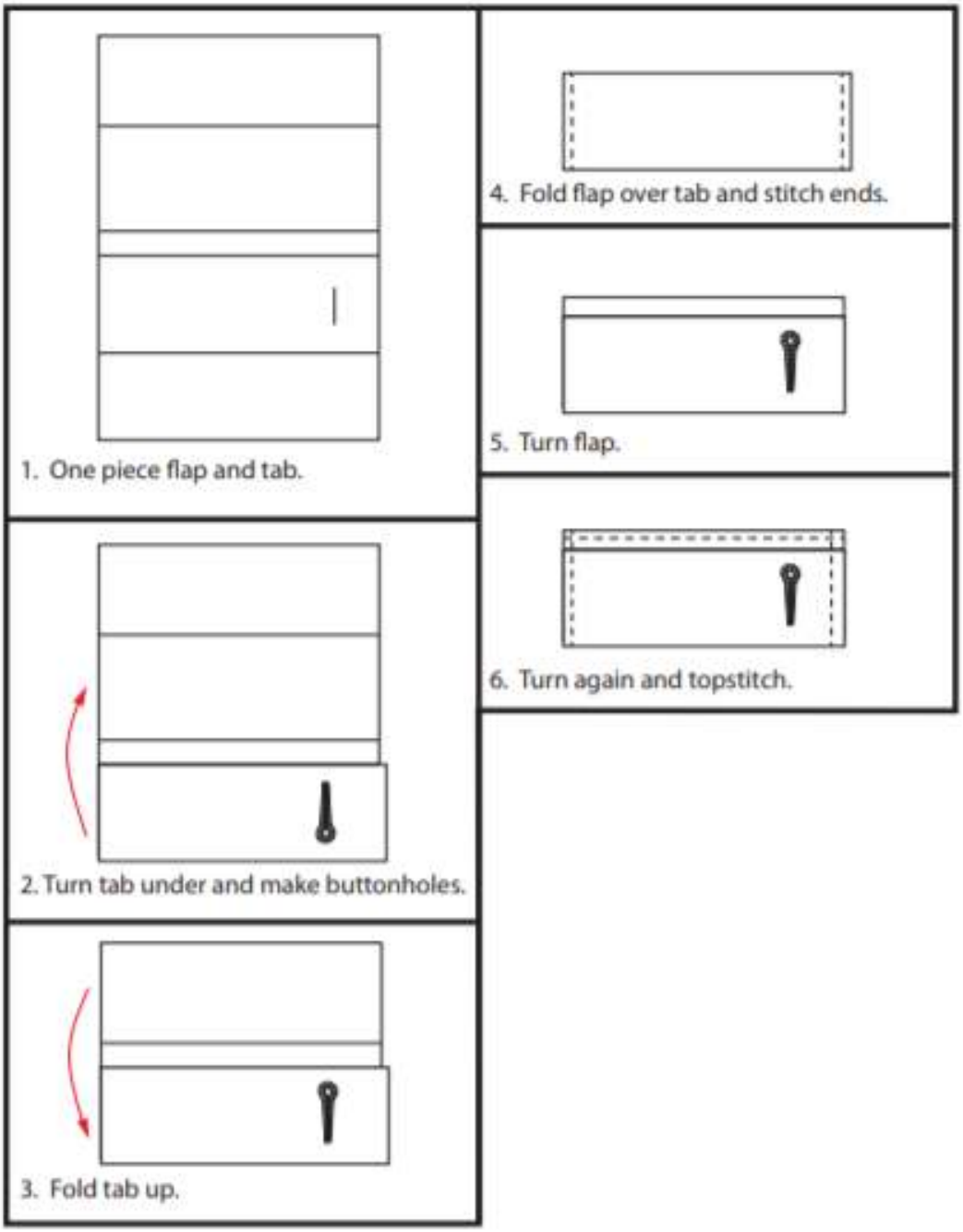


FIGURE 7. Cargo pocket flap construction.

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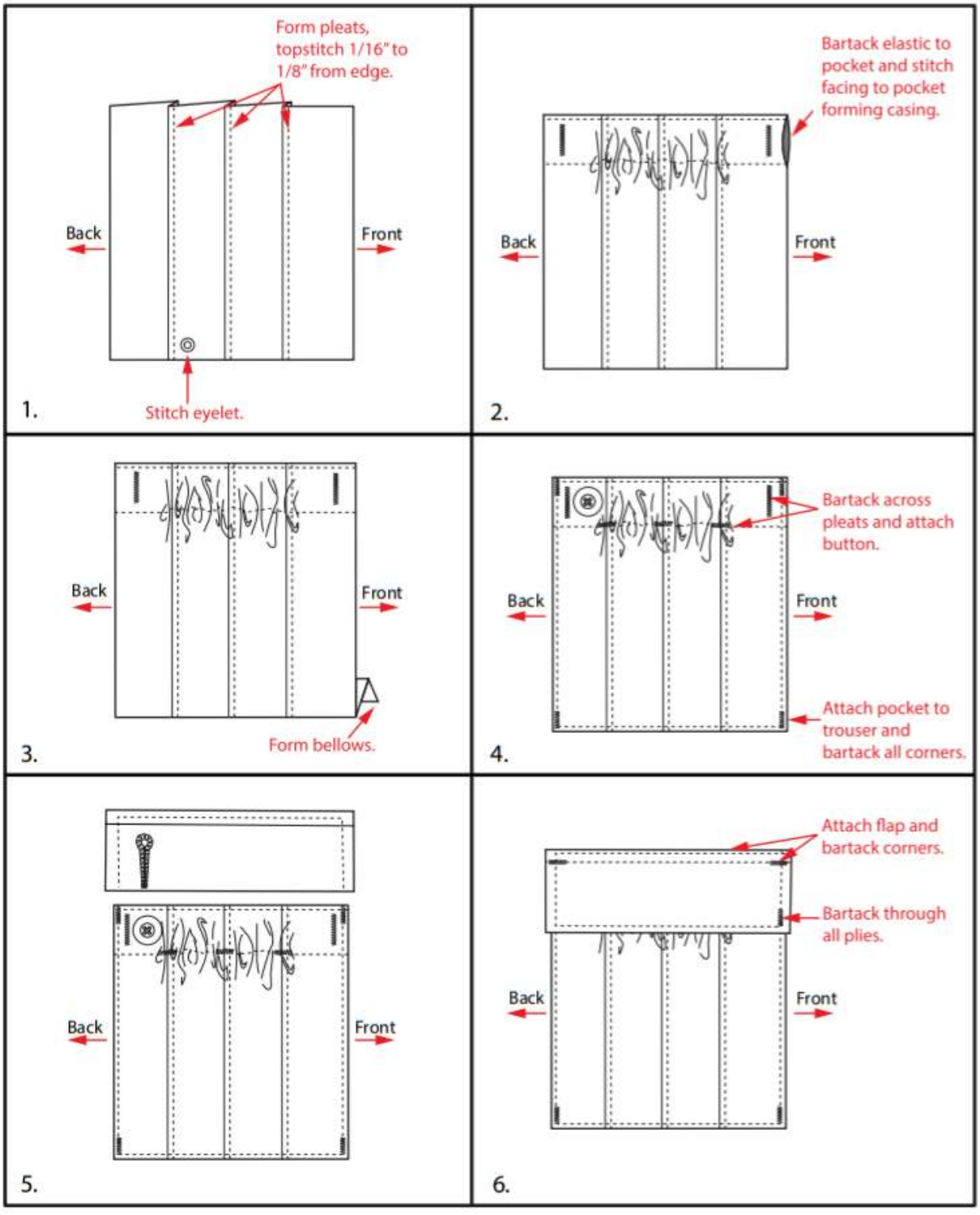


FIGURE 8. Cargo pocket construction.

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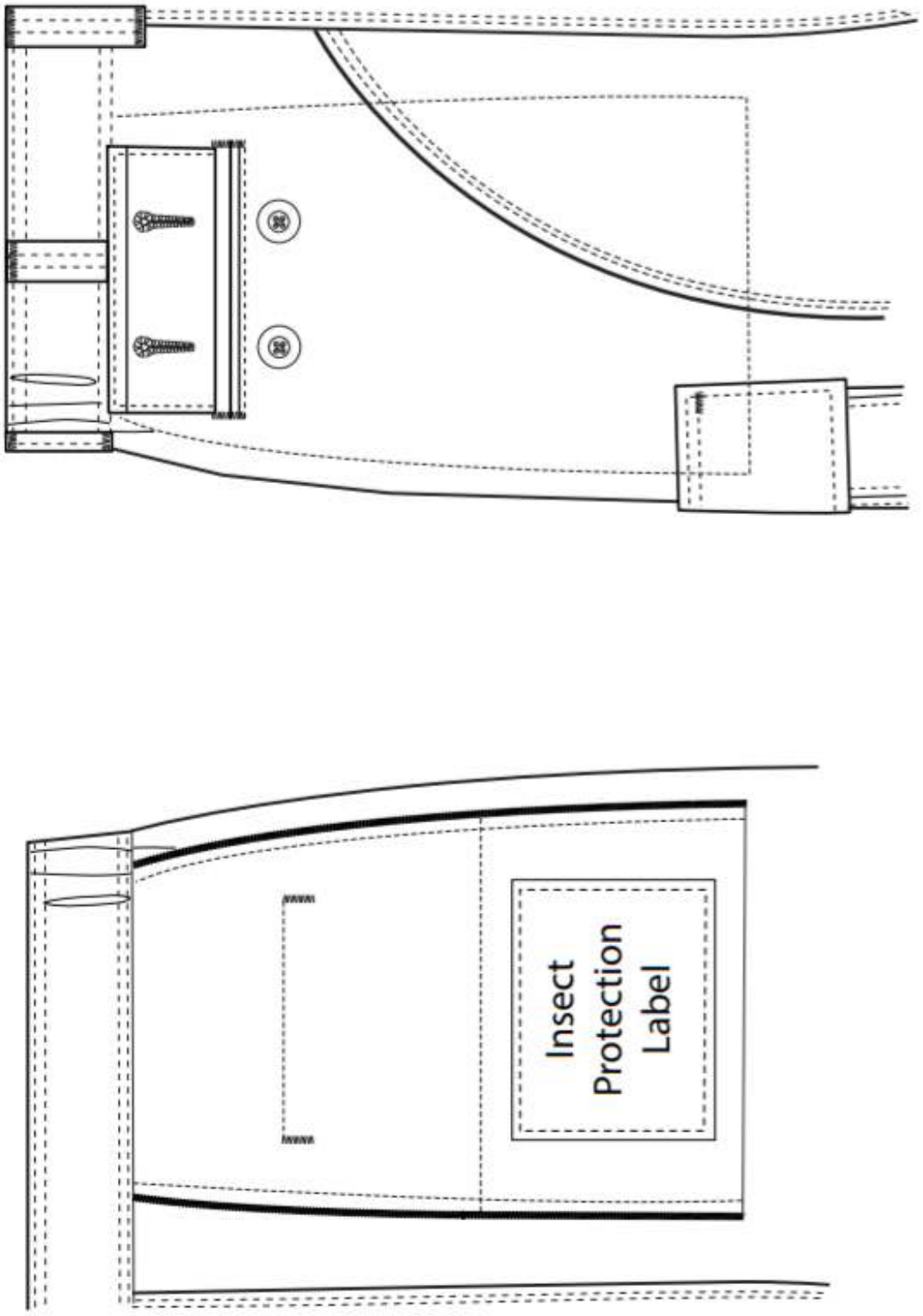


FIGURE 9. Hip pocket – inside and outside.

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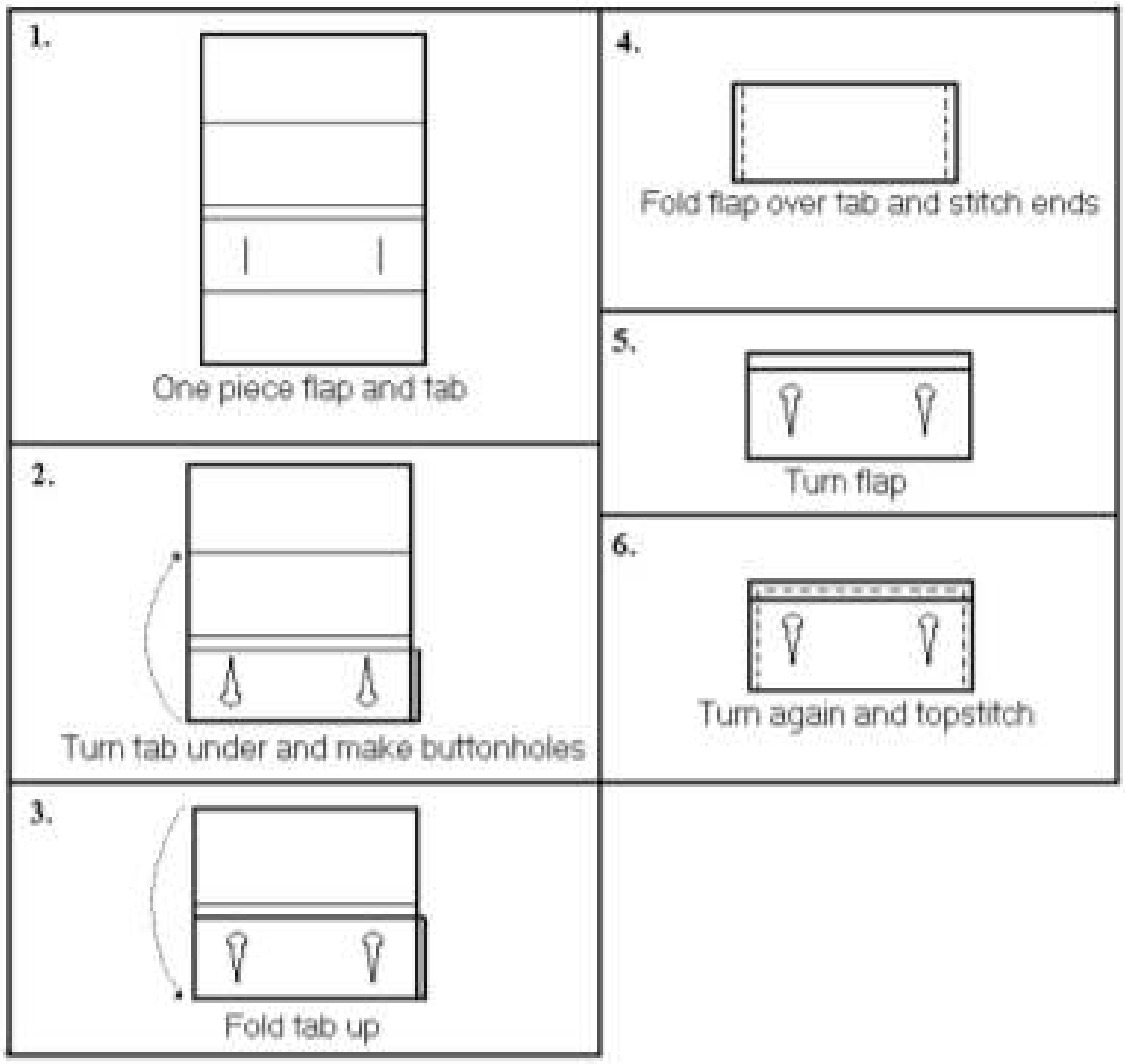
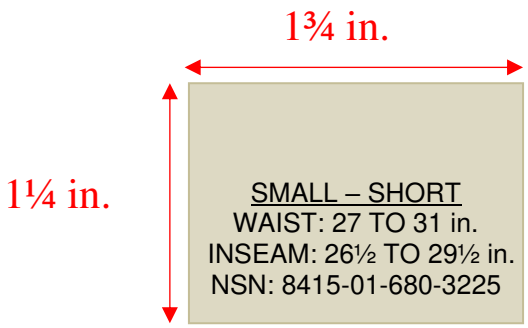


FIGURE 10. Hip pocket flap construction.

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SIZE-LENGTH
WAIST BODY MEASUREMENT
INSEAM BODY MEASUREMENT
NATIONAL STOCK NUMBER

FORMAT



EXAMPLE

FIGURE 11. Size label inscription.

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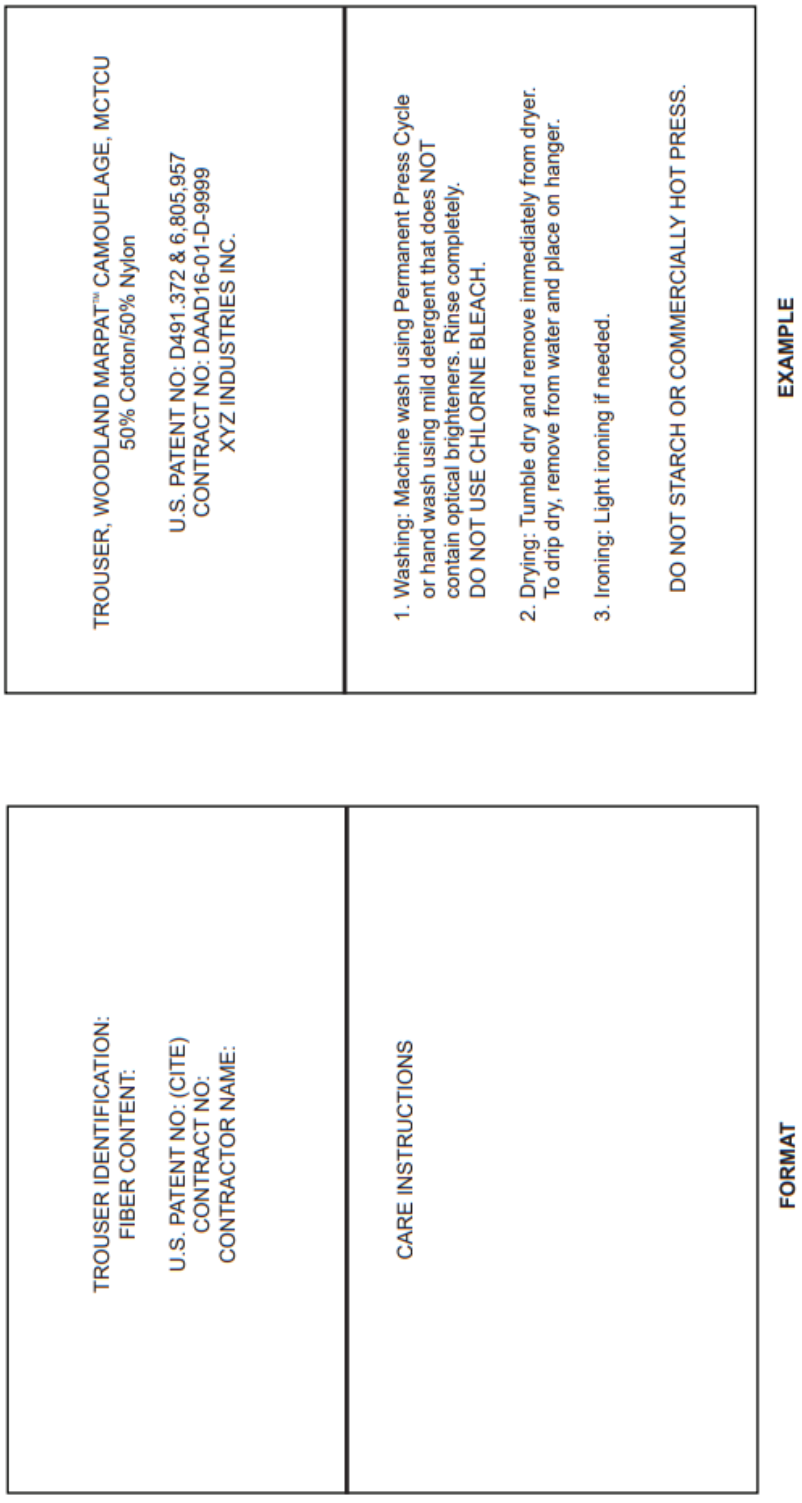


FIGURE 12. Combination label inscription.

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review activities:
Army – AV, CR, MI
Navy – AS, CG1, NU
Air Force – 03, 70
DLA – CT

Civil agency:
GSA – FAS

Preparing activity:
Navy – MC
(Project 8415-2021-055)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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